ECE 443/518 – Computer Cyber Security
Lecture 22 Garbled Circuit

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November 4, 2026

Outline

Garbled NAND

Garbled Circuit

Reading Assignment

- This lecture: Garbled Circuit
- Next lecture (11/9): ICS 2,4,5
  - No in person or zoom lecture due to institutional activities.
  - I'll post a recorded lecture soon.

Outline

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Encrypting Wires

- Use 5 bits for each wire.

Wire	Selection Bit	0	1
O	$S_O = 1$	$O_0 = 10001 = 17$	$O_1 = 00101 = 5$
A	$S_A = 0$	$A_0 = 00110 = 6$	$A_1 = 10000 = 16$
B	$S_B = 1$	$B_0 = 10010 = 18$	$B_1 = 00010 = 2$

- Alice cannot send Bob the above table.
- However, for the computation to proceed, Bob need to know A_a and B_b, and then calculate O_f.
- In general, we should assume Bob has no knowledge of a, b and f so that the idea will work for more complex circuits as multiple levels of gates.

Garbled Truth Table

S(A)	S(B)	E(O)
$S_A = 0$	$S_B = 1$	$e_{A_0,B_0}(O_1) = 6 + 18 + 5 \bmod 32 = 29$
$S_A = 0$	$1 - S_B = 0$	$e_{A_0,B_1}(O_1) = 6 + 2 + 5 \bmod 32 = 13$
$1 - S_A = 1$	$S_B = 1$	$e_{A_1,B_0}(O_1) = 16 + 18 + 5 \bmod 32 = 7$
$1 - S_A = 1$	$1 - S_B = 0$	$e_{A_1,B_1}(O_0) = 16 + 2 + 17 \bmod 32 = 3$

- With the help of an encryption function $e()$, Alice encrypts every gate truth table.
  - e will take A and B as key and O as the plaintext.
  - Subscripts are the actual boolean values, e.g. for A_0 and B_0, we should use O_1 because 0 NAND 0 = 1.
  - Let's use $e_{A||B}(O) = A + B + O \bmod 32$ for our example.

Evaluating Garbled Truth Table

S(A)	S(B)	E(O)
0	1	29
0	0	13
1	1	7
1	0	3

- Alice sends the encrypted truth table to Bob.
  - Hide the binary strings and the selection bits for wires.
- Bob decrypts with this table to obtain O_f from A_a and B_b.
  - Using the first bit of A_a and B_b to identify the row for $E(O_f)$.
  - Since $e_{A||B}(O) = A + B + O \bmod 32$, $O_f = E(O_f) - A_a - B_b \bmod 32$
- For example, for $A_a = 16$ and $B_b = 18$,
  - $S(A_a) = 1$ and $S(B_b) = 1$, so use the third row $E(O_f) = 7$.
  - $O_f = 7 - 16 - 18 \bmod 32 = 5$.
- But Bob can learn S_A and S_B from the table and know what A and B represent

Reordering Tables

- Alice sorts the rows into $S(A)S(B) = 00, 01, 10, 11$.

S(A)	S(B)	E(O)
0	0	13
0	1	29
1	0	3
1	1	7

- Consider $A_a = 16$ and $B_b = 18$ again,
  - Bob still has $S(A_a) = 1$ and $S(B_b) = 1$
  - Now it is the fourth row $E(O_f) = 7$.
  - Bob still computes $O_f = 7 - 16 - 18 \bmod 32 = 5$.
  - Though Bob has no idea what a, b and f are.

Input and Output

- For input wires,
  - Alice sends Bob A_a.
  - Alice uses OT to send Bob B_b.
    - Obviously Bob doesn't want Alice to know b.
- Once Bob calculates O_f, Alice tells what is f.
- Alice has no need to send Bob A_{1-a}.
- Could Alice also send Bob B_{1-b} to avoid using OT?
  - Alice cannot send Bob B_{1-b}.
  - Otherwise Bob can compute $O_{f'}$ from A_a and B_{1-b} and then $a = O_{f'} \oplus O_f$ since $f' = NAND(a, 1 - b)$.
  - In other words, Alice should prevent Bob to evaluate the garbled circuit multiple times using different secrets from Bob.

Outline

Garbled NAND

Garbled Circuit

A More Complicated Circuit

- What about more complicated circuits?
  - E.g. $f = NAND(NAND(a, b), NAND(c, d))$ where Alice provides a and c while Bob provides b and d.
- Identify wires and gates before encrypting them.
  - Wires: A, B, C, D, X, Y, Z
  - Gate 1: $X = NAND(A, B)$
  - Gate 2: $Y = NAND(C, D)$
  - Gate 3: $Z = NAND(X, Y)$

The Garbler Alice: Encrypting Wires

Wire	Selection Bit	0	1
A	$S_A = 0$	$A_0 = 00110 = 6$	$A_1 = 10000 = 16$
B	$S_B = 1$	$B_0 = 10010 = 18$	$B_1 = 00010 = 2$
C	$S_C = 1$	$C_0 = 10100 = 20$	$C_1 = 00001 = 1$
D	$S_D = 1$	$D_0 = 11001 = 25$	$D_1 = 00111 = 7$
X	$S_X = 0$	$X_0 = 00111 = 7$	$X_1 = 11111 = 31$
Y	$S_Y = 0$	$Y_0 = 00000 = 0$	$Y_1 = 10101 = 21$
Z	$S_Z = 1$	$Z_0 = 10001 = 17$	$Z_1 = 00101 = 5$

The Garbler Alice: Encrypting Truth Tables

Gate 1		
S(A)	S(B)	E(X)
$S_A = 0$	$S_B = 1$	$e_{A_0,B_0}(X_1) = 6 + 18 + 31 \bmod 32 = 23$
$S_A = 0$	$1 - S_B = 0$	$e_{A_0,B_1}(X_1) = 6 + 2 + 31 \bmod 32 = 7$
$1 - S_A = 1$	$S_B = 1$	$e_{A_1,B_0}(X_1) = 16 + 18 + 31 \bmod 32 = 1$
$1 - S_A = 1$	$1 - S_B = 0$	$e_{A_1,B_1}(X_0) = 16 + 2 + 7 \bmod 32 = 25$

Gate 2		
S(C)	S(D)	E(Y)
$S_C = 1$	$S_D = 1$	$e_{C_0,D_0}(Y_1) = 20 + 25 + 21 \bmod 32 = 2$
$S_C = 1$	$1 - S_D = 0$	$e_{C_0,D_1}(Y_1) = 20 + 7 + 21 \bmod 32 = 16$
$1 - S_C = 0$	$S_D = 1$	$e_{C_1,D_0}(Y_1) = 1 + 25 + 21 \bmod 32 = 15$
$1 - S_C = 0$	$1 - S_D = 0$	$e_{C_1,D_1}(Y_0) = 1 + 7 + 0 \bmod 32 = 8$

Gate 3		
S(X)	S(Y)	E(Z)
$S_X = 0$	$S_Y = 0$	$e_{X_0,Y_0}(Z_1) = 7 + 0 + 5 \bmod 32 = 12$
$S_X = 0$	$1 - S_Y = 1$	$e_{X_0,Y_1}(Z_1) = 7 + 21 + 5 \bmod 32 = 1$
$1 - S_X = 1$	$S_Y = 0$	$e_{X_1,Y_0}(Z_1) = 31 + 0 + 5 \bmod 32 = 4$
$1 - S_X = 1$	$1 - S_Y = 1$	$e_{X_1,Y_1}(Z_0) = 31 + 21 + 17 \bmod 32 = 5$

The Evaluator Bob

- The garbled circuit sent by Alice

Gate 1		Gate 2		Gate 3	
S(A) S(B)	E(X)	S(C) S(D)	E(Y)	S(X) S(Y)	E(Z)
0 0	7	0 0	8	0 0	12
0 1	23	0 1	15	0 1	1
1 0	25	1 0	16	1 0	4
1 1	1	1 1	2	1 1	5

- Alice sends her inputs: $A_a = 16$, $C_c = 20$
- Alice sends Bob's inputs via OT: $B_1 = 2$, $D_1 = 7$
- Bob's calculation
  - $X_x = 25 - 16 - 2 \bmod 32 = 7$
  - $Y_y = 16 - 20 - 7 \bmod 32 = 21$
  - $Z_z = 1 - 7 - 21 \bmod 32 = 5$
- After Bob shares $Z_z = 5$ with Alice, both party learn the result $f = 1$.

Discussions

- The mechanism works with arbitrary number of NAND gates, and thus any combinational circuits.
  - Bob can evaluate each gate following the topological ordering, without knowing what each gate inputs and gate output mean.
- Overall, there is constant amount of computation and communication per each NAND gate.
  - Efficient in theory.
- A lot of ongoing research to improve its practical performance